

Linear Algebra – Vectors

In linear algebra, a **vector** is an ordered list of numbers (e.g., $\langle v_1, v_2, \dots, v_n \rangle$) representing a quantity with **magnitude** and **direction**, and it can be added to other vectors or scaled by real numbers.

For example, suppose we describe a house using:

Area = **1500** sq ft

Bedrooms = **3**

Age = **5** years

We can represent it as:

$$x = \begin{bmatrix} 1500 \\ 3 \\ 5 \end{bmatrix}$$

This is a **3-dimensional** vector.

Instead of handling each value separately, we can represent the entire object as one mathematical entity.

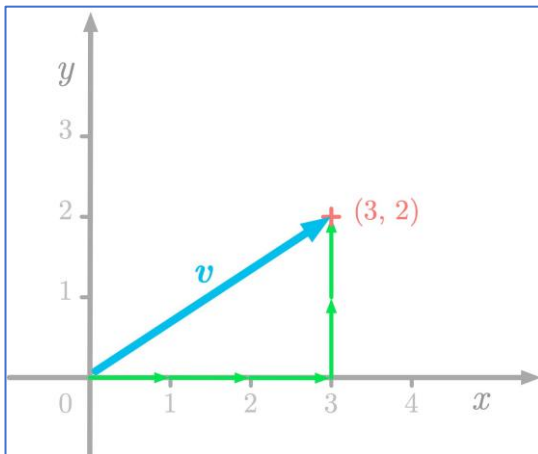
Linear Algebra – Vectors

Vector Intuition

Imagine a simple **2D vector**:

$$v = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

It can represent a point (3,2) or an arrow from the **origin to that point**.

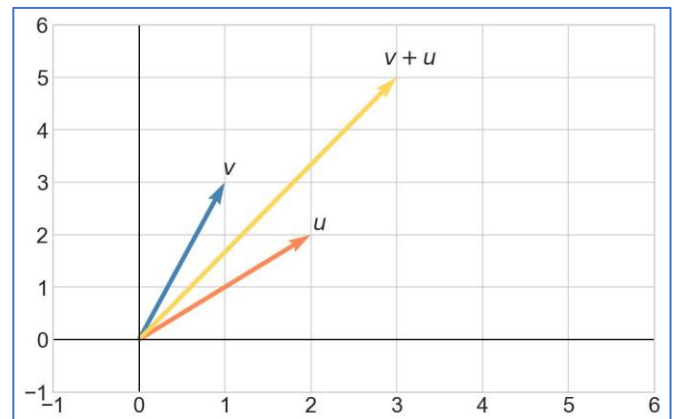


The two numbers tell us its position in the coordinate system.

Vector Addition :

Example:

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$



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Vector Operations

Suppose:

$$\mathbf{A} = [2, 3] \text{ and } \mathbf{B} = [4, 1]$$

Addition : $\mathbf{A} + \mathbf{B} = [6, 4]$

Scalar Multiplication : $3\mathbf{A} = [6, 9]$
(each value multiplied by 3)

Dot Product : $\mathbf{A} \cdot \mathbf{B} = (2)(4) + (3)(1) = 11$

Magnitude : $|\mathbf{A}| = \sqrt{2^2 + 3^2} = \sqrt{13}$

Each operation gives us a different way to manipulate or compare information.

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Why AI Needs Vectors?

This is where the concept becomes interesting.

AI needs a mathematical way to represent **real-world information**.

House **[1500,3,5]**

Customer **[25,12000,8]**

Image **[pixel1,pixel2,pixel3,...,pixeln]**

Word (embedding)
[0.21,-0.43,0.72,...]

Words, images, customers, products, etc.
can all be represented as vectors so
mathematical models can work with them.

Vectors are the mathematical representation
that allows AI systems to manipulate,
compare, and learn from information.